



PROJECT 13

HOUSING PRICE

Non Linear Transformation

PRESENTED BY MUNAWAR

TOPIC REVIEW - NON LINEAR TRANSFORMATION

Non Linear Transformation is the process of changing the value of a variable using a mathematical function that does not form a straight line when graphed.

In this case, we used Log transformation. Log transformation is a way to change data that has very large numbers, very small numbers or a skewed shape. It works by taking the logarithm of each number in the data which helps to “compress” the large values and spread out the small ones.

We use log transformation when:

- Some numbers are extremely large compared to others.
- Data is not following a normal (bell-shaped) distribution.
- We want to make patterns easier to spot for analysis.

source: [geeksforgeeks.org](https://www.geeksforgeeks.org/)

TYPE OF LOG TRANSFORMATION

These three equations represent different types of logarithmic transformations used in regression analysis to model non-linear relationships between variables

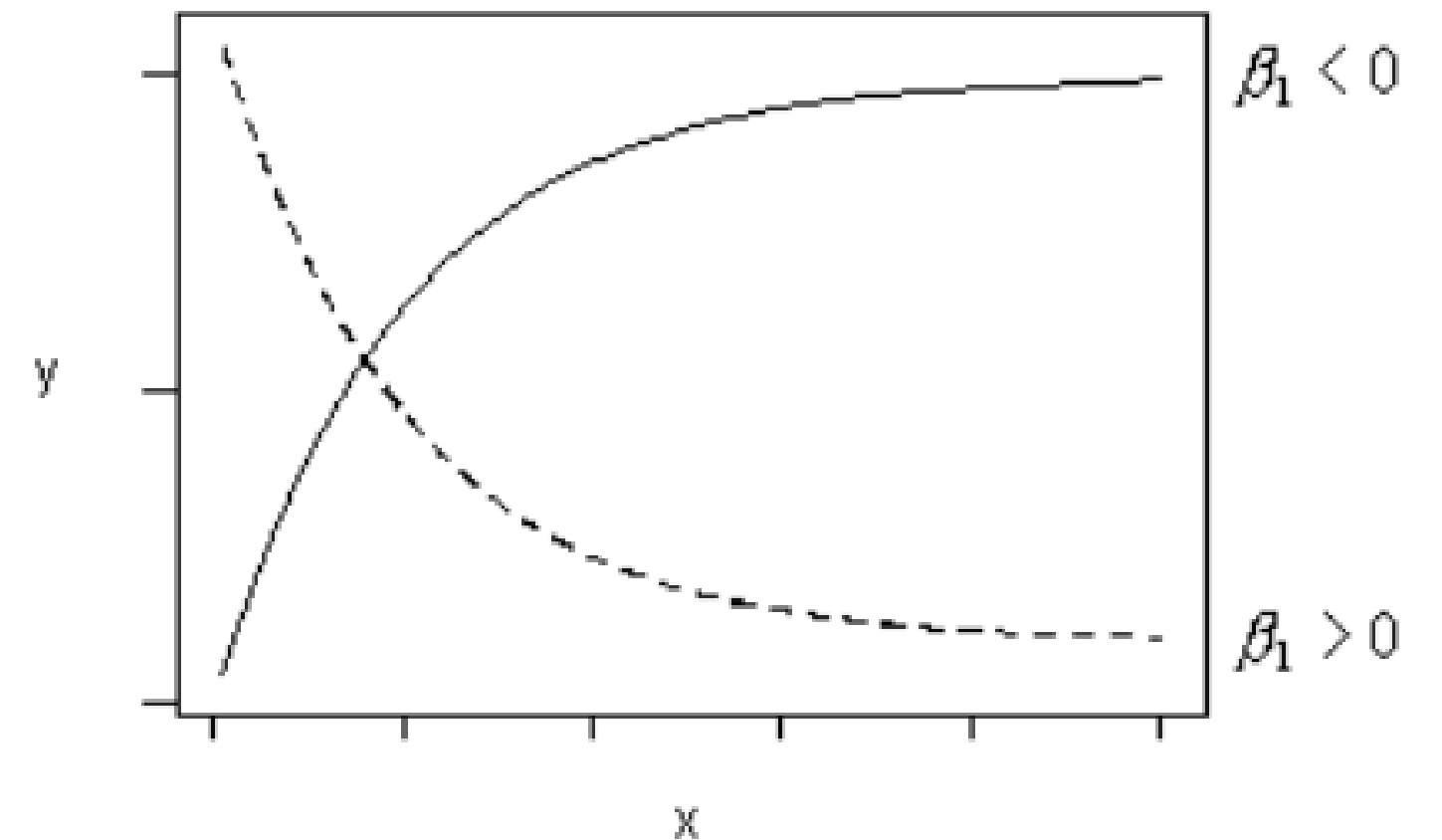
- $y = a + b \log(x)$
 - **Usage:** Used when a change in the percentage of x is associated with an absolute change in y .
 - **Interpretation:** A 1% increase in x is associated with a $b/100$ unit change in y .
- $\log(y) = a + bx$
 - **Usage:** Used when an absolute change in x is associated with a percentage change in y .
 - **Interpretation:** A 1 unit increase in x is associated with a $(b \times 100\%)$ change in y
- $\log(y) = a + b \log(x)$
 - **Usage:** Used to model constant elasticity.
 - **Interpretation:** A 1% increase in x is associated with a $b\%$ change in y . Here, b represents the **elasticity** of y with respect to x .

TYPE OF LOG TRANSFORMATION

We can choose which log transformation to use based on visualization

If you create a scatterplot between the independent variable x and the dependent variable (x vs y), and your data points form a curved pattern like one of the lines in the image, then the best solution for your data to be analyzed with linear regression is to transformation on predictor x

$$y = a + b \log(x)$$

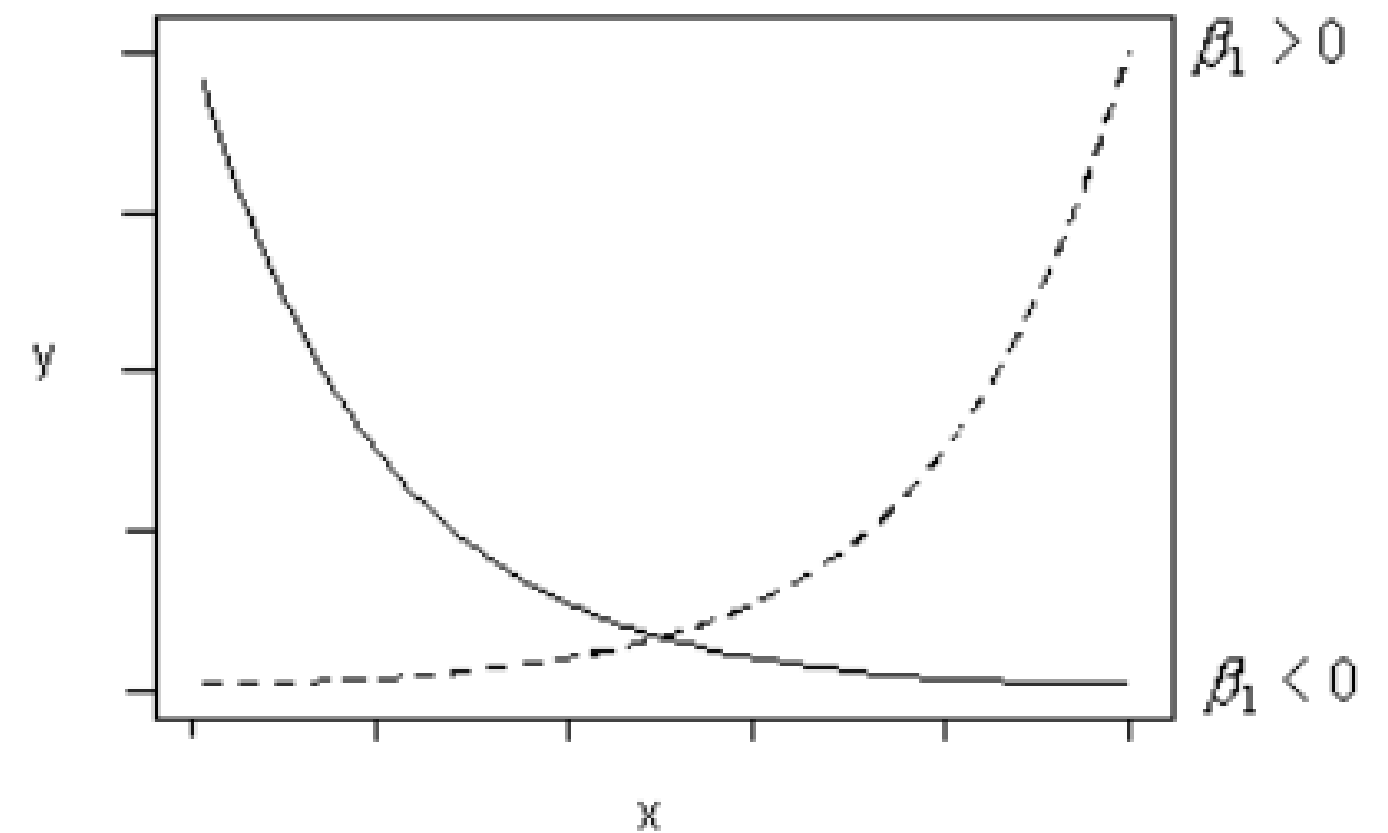


TYPE OF LOG TRANSFORMATION

We can choose which log transformation to use based on visualization

If you create a scatterplot between the original x and y , and then find that your data forms an exponential curve (curving sharply upwards like the letter J, or sliding extremely downwards), then the transformation must be done on the dependent or target variable outcome y .

$$\log(y) = a + bx$$

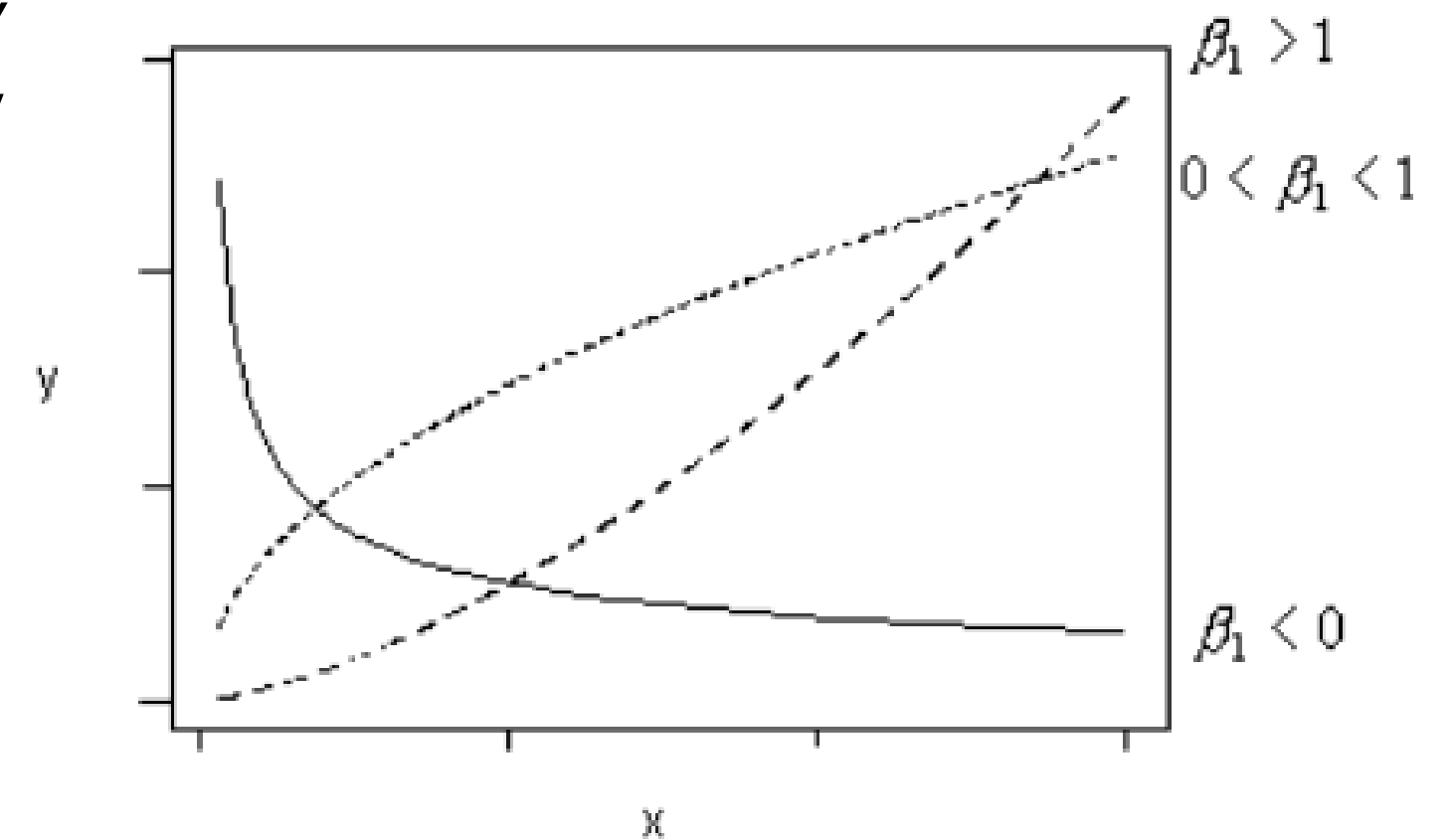


TYPE OF LOG TRANSFORMATION

We can choose which log transformation to use based on visualization

If you plot a scatterplot between the original x and the original y and find that your data form power curves—either exponentially rising or asymptotically falling—then the solution is to transform both variables, both x and y , into logarithms.

$$\log(y) = a + b \log(x)$$



BACKGROUND

Problem

Developers rely on intuition or market averages to set prices — no structured analysis.

This leads to

- Prices set too HIGH → buyers walk away
- Prices set too LOW → profit margin lost

Solution

Use a price elasticity-based approach with log-log OLS regression.

Tasks

- Estimate elasticity via log-log regression
- Calculate optimal price from elasticity
- Simulate profit for area scenarios

METHODOLOGY

In this case, I applied a Log-Log OLS regression model, transforming both the predictor (X) and outcome variable (Y) to estimate the relationships.

Raw Data



01

Price vs Area shows non-linear relationship — simple linear regression is not appropriate

OLS Regression



03

$\log\text{Price} = \alpha + \beta \times \log\text{Area}$
→ coefficient β = price elasticity

Log Transform



02

Apply log to both Price and Area to convert to percentage-change units

Optimal Price



04

Optimal Price = Cost ×
(Elasticity / (1 + Elasticity))

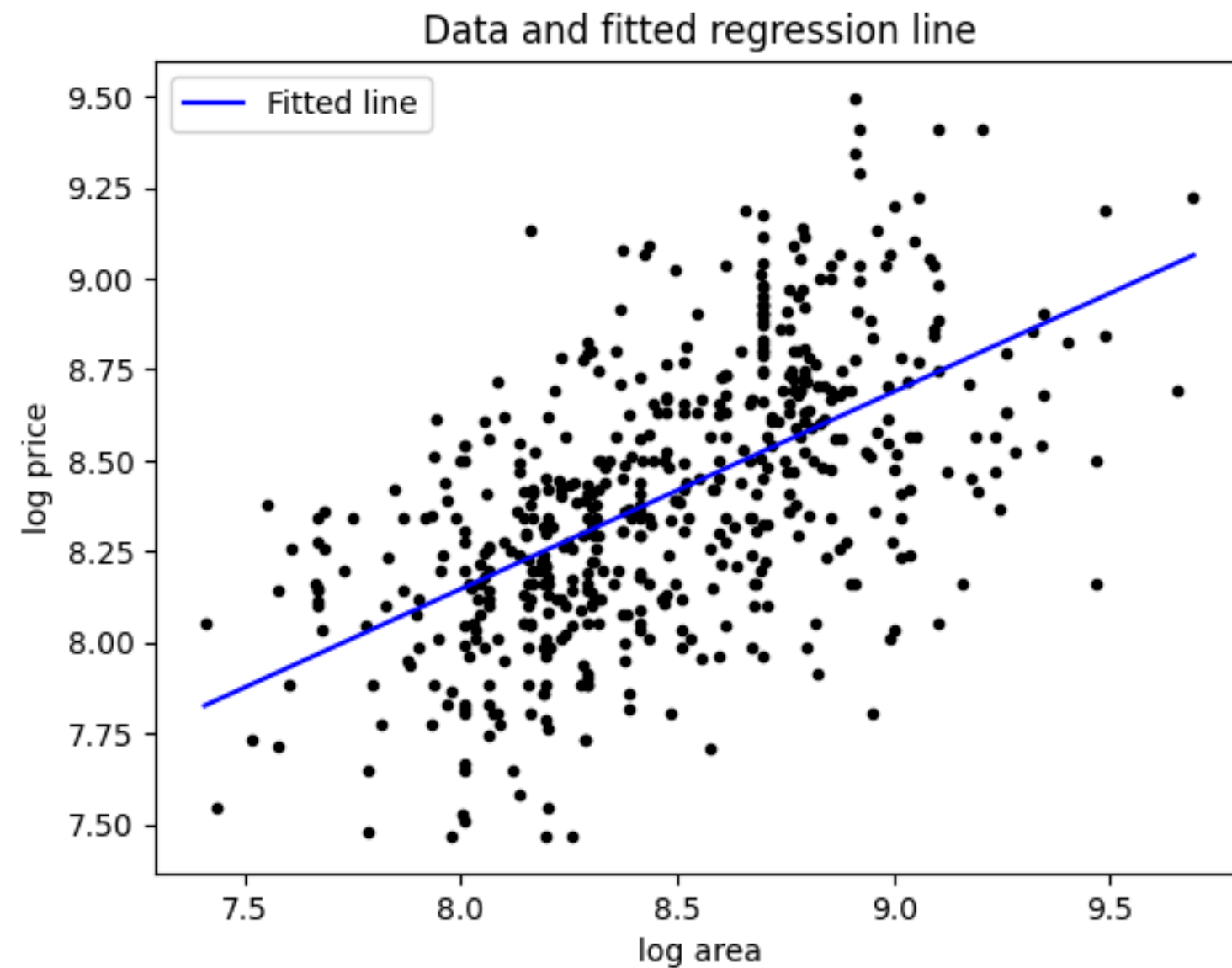
REGRESSION MODEL RESULTS

$logPrice = 3.8 + 0.54 \times logArea$

Parameter	Coefficient	Std Error	Interpretation
Intercept (α)	3.80	0.13	Base log price
logArea (β)	0.54	0.016	Elasticity coefficient

Elasticity 0.54 < 1.0 → Inelastic	Market Type Inelastic Price ↑ won't kill demand	Implication Pricing Power Margins can be expanded
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REGRESSION MODEL RESULTS



$$R^2 \approx 0.3364$$

After logarithmic transformation, the relationship between **logArea** and **logPrice** appears linear — this confirms that the log-log model is appropriate for this data.

OPTIMAL PRICE CALCULATION

$$\text{Optimal Price} = \$10 \times \left(\frac{0.54}{1 + 0.54} \right) = \$3.5 \text{ per unit area}$$

Assumptions

- Cost of capital = \$1 per sq ft (example)
- Elasticity = 0.54 (from OLS model)
- Profit-maximizing price point

OPTIMAL PRICE

\$3.5

per unit area

Profit-maximizing price point
based on inelastic market
conditions

PROFIT SIMULATION BY AREA

THIN MARGIN	MID MARGIN	BEST PROFIT
4,000 sq ft	2,300 sq ft	1,500 sq ft
Market Price: \$4,047K	Market Price: \$2,600K	Market Price: \$2,378K
Capital Cost: \$4,000K	Capital Cost: \$2,300K	Capital Cost: \$1,500K
Profit: \$47K	Profit: \$300K	Profit: \$878K
(1.2% margin)	(13% margin)	(58% margin)

SUMMARY & RECOMMENDATIONS

Key Findings

- Elasticity = 0.54 → inelastic market
- Optimal price ≈ \$3500 per unit area
- 1,500 sq ft yields highest profit (\$878K)
- 4,000 sq ft returns only \$47K profit

RECOMMENDATIONS

- **Target mid-size properties**
Focus on 1,500–2,500 sq ft for best profit-to-cost ratios
- **Price above cost of capital**
Inelastic market allows pricing flexibility without demand loss
- **Adopt data-driven pricing**
Replace heuristic methods with elasticity-based models
- **Expand model predictors**
Add bedrooms, location & amenities for more accurate forecasting

**THANK YOU FOR
YOUR TIME**



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DATA ANALYST PRESENTATION
